

1.

Let  $B$  be a Bloom filter on 11 bits. We will use 3 hash functions:

$$h_1(x) = x \bmod 11$$

$$h_2(x) = 10x - 1 \bmod 11$$

$$h_3(x) = \lfloor x/11 \rfloor \bmod 11$$

1. Insert the elements listed in the table into the Bloom filter. The columns are indexed as seen in the first row. Start from the empty filter (all 0s) and write 0/1 as the next element is added to the filter. By the end of this exercise your filter should hold 6 elements.

| index | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|-------|---|---|---|---|---|---|---|---|---|---|----|
| 10    |   |   |   |   |   |   |   |   |   |   |    |
| 123   |   |   |   |   |   |   |   |   |   |   |    |
| 2     |   |   |   |   |   |   |   |   |   |   |    |
| 0     |   |   |   |   |   |   |   |   |   |   |    |
| 111   |   |   |   |   |   |   |   |   |   |   |    |
| 1000  |   |   |   |   |   |   |   |   |   |   |    |

2. What is returned if we query whether the following numbers are in the Bloom filter (write positive or negative for each): 37, 7, 1100, 124?
3. At this state of the Bloom filter holding the above six elements, what is the probability that a number that is not in the filter returns positive upon look up? Show your computation.

Show transcribed data

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$$0\text{ s}$$

) and write

$$0/1$$

0/1

as the next element is added to the filter. By the end of this exercise your filter should hold 6 elements. 2. What is returned if we query whether the following numbers are in the Bloom filter (write positive or negative for each):

37, 7, 1100, 124

? 3. At this state of the Bloom filter holding the above six elements, what is the probability that a number that is not in the filter returns positive upon look up? Show your computation.

## Expert Answer

This solution was written by a subject matter expert. It's designed to help students like you learn core concepts.

### Step-by-step

... 1st step

≡ All steps

✓ Answer only

Step 1/3



1) I am considering  $h_2(x) = (10x - 1) \bmod 11$

For  $x = 123$

$$h_1(x) = 123 \bmod 11 = 2$$

$$h_2(x) = (123 \times 10 - 1) \bmod 11 = 8$$

$$h_3(x) = \left( \text{floor of } \left( \frac{123}{11} \right) \right) \bmod 11 = 11 \bmod 11 = 0$$

=> setting bit 2nd, 8th, and 0th to 1

Similarly we calculate for all starting from 10, then 123, 2, 0, 111, then 1000, we obtain

|       |   |   |   |   |   |   |   |   |   |   |    |
|-------|---|---|---|---|---|---|---|---|---|---|----|
| Index | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|-------|---|---|---|---|---|---|---|---|---|---|----|

| Index     | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|-----------|---|---|---|---|---|---|---|---|---|---|----|
| Initially | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  |
| 10        | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1  |
| 123       | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1  |
| 2         | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1  |
| 0         | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1  |
| 111       | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1  |
| 1000      | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1  |

### Step 2/3



2)

for 37 hashing function bit contains 0 => it is definitely not present => Negative

for 7 hashing function bit contains 0 => it is definitely not present => Negative

for 1100, hashing function bits are all 1 => it is probably present => Positive

for 124, hashing function bit contains 0 => it is definitely not present => Negative

### Step 3/3



Formula Used :  $P = \text{Probability of false positive} = \left[1 - \left(1 - \frac{1}{m}\right)^{kn}\right]^k$

m: size of bit array => m = 11

k: Number of hashing functions => k = 3

n: number of elements that array is holding or expecting => n = 6

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$$P = \left[1 - \left(1 - \frac{1}{11}\right)^{3 \times 6}\right]^3 = \left[1 - (0.909090909)^{3 \times 6}\right]^3$$

$$\Rightarrow P = [0.82014121]^3 = 0.5516529$$

### Final answer



Probability of false positive = 0.5516529

2.

1. Suppose we have two Bloom filters A and B with the same number of bits and using the exact same hash functions. A is representing the set X (the elements  $x \in X$  are stored in A) and B is representing Y. Answer the following questions

(a) Let  $C = A \wedge B$  be the Bloom filter formed by computing the bitwise Boolean and operation between A and B. Prove that C is not the same as the Bloom filter that would be constructed by adding the elements of the set  $X \cap Y$  one at a time.

(b) Does C correctly represent the set  $X \cap Y$ , in the sense that it gives a positive answer for membership queries of all elements in this set? Explain why or why not.

(c) Let  $D = A \vee B$  be the Bloom filter formed by computing the bitwise Boolean or (inclusive) operation between A and B. Show that D does represent the union of the sets X and Y.

2. Suppose the size m (number of bits) of the Bloom filter B is a power of 2 and B represents the elements in set X. Now we want to halve the size of B. Create the Bloom filter B0 on  $m/2$  bits by taking the bitwise Boolean or of the first and second half of the bit vector of B. Thus, if we denote the first  $m/2$  bits of B by B1 and the second  $m/2$  bits by B2, then  $B0 = B1 \vee B2$ .

Suppose  $h_1(x), \dots, h_r(x)$  are the hash functions originally used by B. To look up an element  $x \in X$  in B0 we compute for each  $i$   $h_i(x) \bmod (m/2)$ .

Show that B0 is still representing X. That is, show that for every element  $x \in X$  the look up in B0 will return positive.

## Expert Answer

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## Bloom Filters:

Bloom filters is probabilistic and space efficient data structure used to test element present in or not in set.

Bloom filters has a bit array of  $m$  size that all set to zero (0, false). here  $k$  hash functions that give the position in bit array to set 1.

Let set  $S = \{11, 5, 3\}$  and  $T = \{3, 6\}$   $k=2$  then

$$h_1(x) = x \bmod 5 \text{ and } h_2(x) = (3x+5) \bmod 7$$

Bloom filter  $B_S$  and  $B_T$  of size 7 that is  $m=7$ .

$B_S$ :

initial  $B_S$  is 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|

add element 11 to  $B_S$   $h_1(11) = 11 \bmod 7 = 4$

$$h_2(11) = 3 \times 11 + 5 \bmod 7 = 3$$

so and on 5, 3 we will get

$B_S =$ 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|

$B_T$ : After add set  $T$  element in  $B_T$  then

$B_T =$ 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|

a) Bitwise OR of  $B_S, B_T$  is  $B =$ 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|

and union of set  $S$  and  $T$  is  $S \cup T = \{11, 5, 3, 6\}$  then find Bloom filters of  $B_{S \cup T}$

that is prove that bitwise OR of 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|

$B_S, B_T$  is  $B_{S \cup T}$

b) bitwise AND of  $B_S$  and  $B_T$  is  $B =$ 

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|

and Intersection of  $S \cap T = \{3\}$  that is,  $B_{S \cap T} =$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|

here  $B$  and  $B_{S \cap T}$  not same.

Prove that bitwise AND of  $B_S, B_T$  might not produce  $B_{S \cap T}$

1. There are  $n$  socks in the drawer, some of which are red and the remaining socks are blue. (There is no restriction on having odd or even number of each color.) We draw a sock from the drawer, each sock is equally likely to be selected. If the sock is red, we throw it out and replace it with a blue sock, if it is blue then we put it back. Repeat this process many times. (a) Describe how to model this exercise using a Markov chain. Explain what the states  $S_i$  correspond to, how many states there are and when can we transition from one state to another.

(b) Show that this Markov chain has the memoryless property (sometimes also called Markov property).

(c) compute what the transition probability  $p_{ij}$  between states  $S_i$  and  $S_j$  is. (Hint: Your answer will depend on how many red/blue socks each state represents.)

(d) Is this Markov chain irreducible and aperiodic, why or why not?

(e) Does this Markov chain have a stationary distributions? If not, then why not? If yes, then what is it?

2. Now we have  $n$  socks, some of which are red the others blue. We again draw socks from the drawer at random. If a sock is red, then we throw it out and replace it with a blue one. If a sock is blue, then we put it back. However, this time we will count how many times in a row we draw a blue sock. Every time that we have blue twice in a row, we throw the second blue sock out and replace it with a red sock.

(a) is this Markov chain memoryless?

(b) Compute the transition probability between states in this Markov chain.

(c) Is this Markov chain irreducible and aperiodic, why or why not?

## Expert Answer

This solution was written by a subject matter expert. It's designed to help students like you learn core concepts.

### Step-by-step

... 1st step

≡ All steps

✓ Answer only

#### Step 1/3



- The solution for the question is given in the next step.
- The solution for the question is given in the detail.

#### Step 2/3



Here is the solution for the given question.

1.

(a) We can model this exercise using a Markov chain with  $n+1$  states, where each state  $S_i$  corresponds to the number of red socks in the drawer. The first state  $S_0$  corresponds to 0 red socks, the second state  $S_1$  corresponds to 1 red sock, and so on until the last state  $S_n$  corresponds to  $n$  red socks. We can transition from one state to another when we draw a sock from the drawer. If the sock is red, we transition from state  $S_i$  to state  $S_{i-1}$ , and if the sock is blue, we transition from state  $S_i$  to state  $S_{i+1}$ . This means that the number of red socks in the drawer is always changing, and the probability of transitioning from one state to another is dependent on the number of red socks in the drawer.

(b) This Markov chain has the memoryless property because the probability of transitioning from one state to another is independent of the history of the chain. Specifically, the probability of transitioning from state  $S_i$  to state  $S_{i+1}$  is always the same regardless of the number of times we have previously drawn a blue sock.

(c) The transition probability  $p_{ij}$  between states  $S_i$  and  $S_j$  is given by:

$$p_{ij} = P(S_i \rightarrow S_j) =$$

```
begin{cases}
\frac{i}{n} & \text{if } j = i+1 \setminus \\
\frac{n-i}{n} & \text{if } j = i-1 \setminus \\
0 & \text{otherwise}
\end{cases}
```

This is because the probability of transitioning from state  $S_i$  to state  $S_{i+1}$  is equal to the probability of drawing a blue sock, which is equal to the number of blue socks in the drawer divided by the total number of socks in the drawer (i.e.  $n-i/n$ ). Similarly, the probability of transitioning from state  $S_i$  to state  $S_{i-1}$  is equal to the probability of drawing a red sock, which is equal to the number of red socks in the drawer divided by the total number of socks in the drawer (i.e.  $i/n$ ).

(d) This Markov chain is irreducible because we can transition from any state to any other state. It is also aperiodic because the transition probability from any state to itself is always greater than 0.

(e) This Markov chain does have a stationary distribution. The stationary distribution is given by:

$$p_i = P(S_i) =$$

```
begin{cases}
\frac{1}{2^n} & \text{if } i = 0,1,2,\dots,n \setminus \\
0 & \text{otherwise}
\end{cases}
```

This is because the probability of being in any particular state is equal to the probability of transitioning from the previous state to that state multiplied by the probability of being in the previous state. Since the transition probabilities are all equal, the probability of being in any particular state is equal to the probability of being in the initial state, which is  $1/2^n$ .

### Step 3/3

2.

(a) Yes, this Markov chain is memoryless because the transition probabilities between states do not depend on the previous states, and only depend on the current state. This means that the probability of transitioning from one state to another does not depend on the history of the chain, and only depends on the current state.

(b) The transition probability between states in this Markov chain is given by:

$$P(\text{blue}|\text{blue}) = 1 - P(\text{red}|\text{blue}) = 1 - P(\text{red}) = 1 - n/(n+1)$$

$$P(\text{red}|\text{blue}) = P(\text{red}) = n/(n+1)$$

The transition probability between states in this Markov chain is given by the probability of drawing a red sock given that the previous sock was blue, which is equal to the probability of drawing a red sock in general, which is  $n/(n+1)$ . The probability of drawing a blue sock given that the previous sock was blue is  $1 - n/(n+1)$ .

(c) Yes, this Markov chain is irreducible and aperiodic because the transition probabilities between states are non-zero for all states, and the states are all accessible from each other. This means that the chain is irreducible, as all states can be reached from any other state, and aperiodic, as the transition probabilities between states are non-zero.

### Final answer

The brief solution is given for the given question.

4.

$G$  is a unknown graph with  $m$  edges. We know the value of  $m$ , but we don't know the number of vertices  $n$ , nor the edges themselves.

We can probe  $G$  using a blackbox sampler as follows: At each step, the sampler picks a random edge  $(u, v)$  among the  $m$  edges, and with probability  $1/2$  chooses  $u$  and returns  $\deg(u)$  or chooses  $v$  and returns  $\deg(v)$ , where  $\deg(u)$  denote the degree of vertex  $u$ . (Note that the sampler doesn't give us the node id, only the degree.)

1. For a given node  $u$ , what is the probability  $p(u)$  that in a call to the sampler it returns the information  $\deg(u)$ ?

2. Define the function  $f : V \rightarrow \mathbb{R}$ , such that  $f(u) = 2m \deg(u)$ . Let  $X$  be the random variable whose values are  $f(u)$  with probability  $p(u)$  for each  $u \in V$ . (We will assume that there are no isolated nodes, i.e.  $\deg(u) \neq 0$ .) Note, that – despite the notation – the definition of  $f(u)$  does not require us to know which node  $u$  the degree  $\deg(u)$  is corresponding to. Compute the expectation  $E[X]$ .



**Step 1/2**

1. The probability that the sampler chooses a given edge  $(u, v)$  and returns  $\deg(u)$  is  $1/2$ . Since  $\deg(u)$  is the degree of vertex  $u$ , and the degree of a vertex is simply the number of edges that are incident to that vertex, the probability that in a call to the sampler it returns the information  $\deg(u)$  is the number of edges that are incident to  $u$  divided by the total number of edges in the graph, which is  $p(u) = \deg(u)/m$ .

**Step 2/2**

2.  
The expected value of a random variable is the sum of the values of the random variable multiplied by their respective probabilities. Since the random variable  $X$  takes on the values  $f(u)$  with probability  $p(u)$ , we can compute the expectation of  $X$  as follows:

$$\begin{aligned} E[X] &= \sum(p(u) * f(u)) \\ &= \sum(p(u) * 2m * \deg(u)) \\ &= \sum(2m * (\deg(u) / m) * \deg(u)) \\ &= \sum(2 * \deg(u)^2) \end{aligned}$$

Since the sum of the degrees of all vertices in a graph is equal to twice the number of edges in the graph, we can simplify the expression further as follows:

$$\begin{aligned} E[X] &= \sum(2 * \deg(u)^2) \\ &= 2 * \sum(\deg(u)^2) \\ &= 2 * \sum(\deg(u))^2 / m \\ &= 2 * (2m)^2 / m \\ &= 8m. \end{aligned}$$

Thus, the expectation of  $X$  is equal to  $8m$ .

**Final answer**

please refer to all above steps